

React Course Enrollment Dashboard

ReactJS Code

```
import React, { useState } from "react";

function CourseEnrollmentDashboard() {
  const [studentsMap, setStudentsMap] = useState(
    new Map([
      [
        1,
        {
          id: 1,
          name: "Alice",
          enrolledCourses: new Set(["Java", "DBMS"]),
          gpa: 8.9,
        },
      ],
      [
        2,
        {
          id: 2,
          name: "Bob",
          enrolledCourses: new Set(["React", "Java"]),
          gpa: 9.2,
        },
      ],
    ])
  );

  const [filterCourse, setFilterCourse] = useState("");

  const addStudent = () => {
    const newStudent = {
      id: 3,
      name: "Charlie",
      enrolledCourses: new Set(["OS", "Java"]),
      gpa: 8.5,
    };

    const updatedMap = new Map(studentsMap);
    updatedMap.set(newStudent.id, newStudent);

    setStudentsMap(updatedMap);
  };

  const removeStudent = (id) => {
    const updatedMap = new Map(studentsMap);
    updatedMap.delete(id);

    setStudentsMap(updatedMap);
  };

  const students = [...studentsMap.values()];

  const sortedStudents = [...students].sort(
    (a, b) => b.gpa - a.gpa
  );

  const uniqueCourses = [
    ...students.reduce((courseSet, student) => {
      student.enrolledCourses.forEach((course) =>
        courseSet.add(course)
      );
      return courseSet;
    }, new Set())
  ];

  const filteredStudents = students.filter((student) =>
    filterCourse === ""
      ? true
      : student.enrolledCourses.has(filterCourse)
  );
}
```

```

return (
  <div>
    <h2>Course Enrollment Dashboard</h2>

    <button onClick={addStudent}>
      Add Student
    </button>

    <input
      type="text"
      placeholder="Filter by Course"
      value={filterCourse}
      onChange={(e) =>
        setFilterCourse(e.target.value)
      }
    />

    <h3>Unique Courses</h3>
    <ul>
      {uniqueCourses.map((course, index) => (
        <li key={index}>{course}</li>
      ))}
    </ul>

    <h3>Students (Sorted by GPA)</h3>
    <ul>
      {sortedStudents.map((student) => (
        <li key={student.id}>
          {student.name} - GPA: {student.gpa}
          <button
            onClick={() =>
              removeStudent(student.id)
            }
          >
            Remove
          </button>
        </li>
      ))}
    </ul>

    <h3>Filtered Students</h3>
    <ul>
      {filteredStudents.map((student) => (
        <li key={student.id}>
          {student.name}
        </li>
      ))}
    </ul>
  </div>
);
}

export default CourseEnrollmentDashboard;

```

Time Complexity Analysis

Filtering Students by Course:

Let n be the number of students. The filter operation checks each student once to determine whether the specified course exists in the student's Set of enrolled courses. Since Set lookup is $O(1)$ on average, the overall time complexity is $O(n)$.